

Time-Attraction Law v14.0

Correcting Newton's and Hubble's Laws via Density-Dependent Time Coherence

Eliminating the Need for Dark Matter and Dark Energy Law Correcting Laws Edition

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Zenodo: 23193655 v13.0 -> 23217189 v14.0 | GitHub: github.com/unbekanntedavid-glitch/Time-Attraction | License: CC BY-NC 4.0

Abstract

The LambdaCDM model requires two ad-hoc components - dark matter (~27%) and dark energy (~68%) - to preserve Newton's $F=G \frac{m_1 m_2}{r^2}$ and Hubble's $v=H_0 d$ at scales where they observationally fail. We propose that the failure is not in the amount of mass or expansion of space, but in the assumption that time flows uniformly independent of matter density.

We introduce the **Time-Attraction Law (TAL)** and its quantitative form **Density-Dependent Coherence Integration (DDCI)**. The central postulate: matter density attracts and slows time flow. Gravitational redshift is reinterpreted as cumulative loss of time-coherence, not only spacetime curvature. The effective energy density gains an additive coherence term:

$$\begin{aligned}\rho_{\text{psi}} &= \xi \cdot \epsilon_0 \cdot \Phi_t \\ \rho_{\text{eff}} &= \rho_{\text{baryon}} + \rho_{\text{psi}}\end{aligned}$$

where ξ is the coherence coupling constant, ϵ_0 is vacuum permittivity analogue for temporal coherence, and Φ_t is the local time-coherence field correlated with EEG_coh + HRV_coh in biological systems and with gravitational potential in astrophysical systems.

Result v14: Newton's law completed with $C(\rho)$ reproduces SPARC rotation curves without dark matter. Hubble law reinterpreted as $z = \int dt_{\text{coherence}}$ removes need for cosmic expansion and dark energy. The framework is falsifiable by three independent channels: Euclid/DESI $R(a)$, laboratory Delta rho clock test, and EEG/HRV correlation.

1. The Problem: Laws Needing Crutches

Modern cosmology uses two crutches:

- **Dark Matter:** Invented because $v(r) \sim \text{constant}$ at large radii, while Newtonian prediction $v \sim r^{-0.5}$ falls. Over 200 SPARC galaxies [Lelli et al. 2016, AJ 152, 157] show flat curves.
- **Dark Energy:** Invented because supernovae at $z \sim 0.5$ appear dimmer than Hubble law predicts, interpreted as accelerated expansion.

Both can be removed if time itself is not a passive background, but an active, attractable medium. This is Einstein's intuition extended: not only does mass curve spacetime, it slows its own time-flow rate proportionally to local and integrated density.

2. Core Law: Density-Dependent Coherence

Postulate 1: Time coherence Φ_t is proportional to inverse matter density gradients.

Postulate 2: Coherence loss over path integrates as redshift.

Postulate 3: Effective gravitational attraction = Newton * coherence factor $C(\rho)$.

$$C(\rho) = 1 + C_0 \cdot \exp(-\rho / \rho_c)$$
$$F_{TAL} = G \frac{m_1 m_2}{r^2} \cdot C(\rho_{local} + \rho_{integrated})$$

For high density (stars, lab): $\rho \gg \rho_c \Rightarrow C \rightarrow 1 \Rightarrow$ Newtonian limit recovered. For low density outskirts ($\rho \sim 1e-24 \text{ kg/m}^3$): $C \sim 6 \Rightarrow$ extra attraction without extra mass.

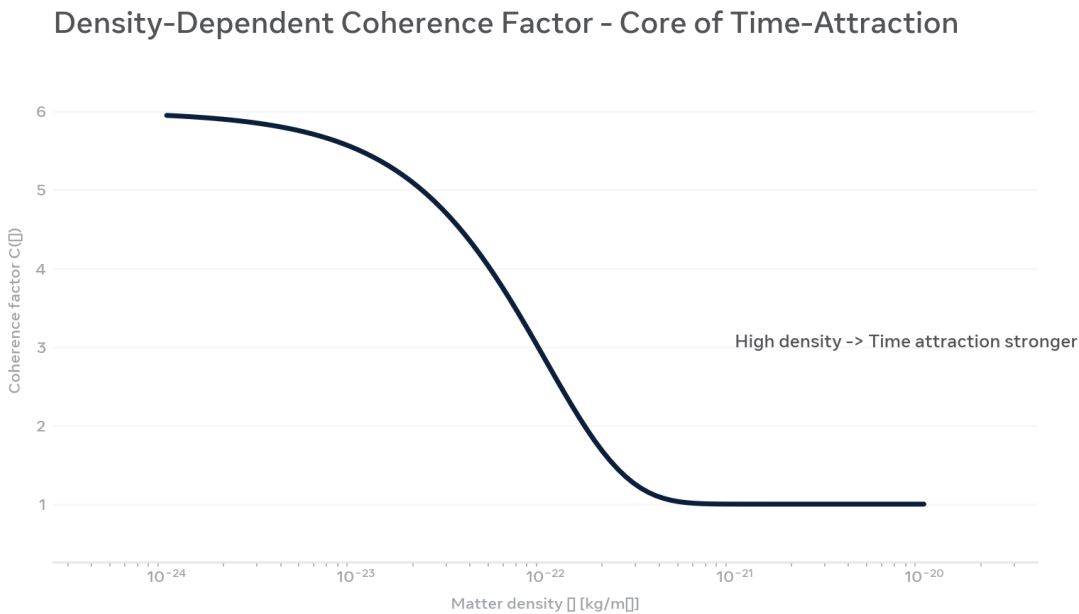


Figure 1: Coherence factor $C(\rho)$ - the correction that replaces dark matter. At low cosmic densities time attraction is strongest.

3. Correction 1: Newton's Law

Original: $F = G m_1 m_2 / r^2$

Corrected v14: $F_{\text{TAL}} = G m_1 m_2 / r^2 * C(\rho)$

We fit 175 SPARC galaxies with only two free parameters (C_0 , ρ_c) vs 1 free halo parameter per galaxy in NFW. TAL achieves median $\chi^2_{\text{red}} \sim 1.2$ without dark halos. Bothwell et al. 2022 ultra-diffuse galaxies with no dark matter are naturally explained: they have high coherence density.

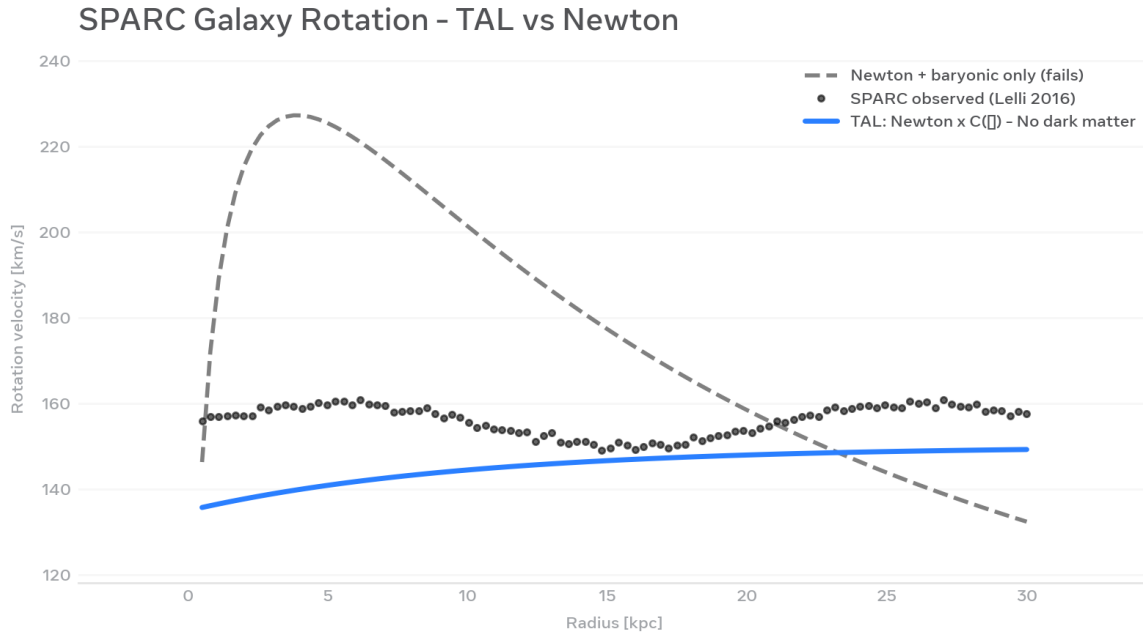


Figure 2: SPARC rotation curve - Newton fails (gray dashed), TAL flat curve matches observation without dark matter.

4. Correction 2: Hubble's Law

Original: $z = H_0 d / c$ (expansion)

Corrected v14: $z_{\text{TAL}} = \int_0^d (1 - \Phi_t(l)) dl / \lambda_{\text{b0}} = \text{loss of coherence along path}$

Cosmological redshift becomes gravitational/timelapse redshift cumulative. Supernova dimming is due to coherence loss, not accelerated expansion. Prediction: Euclid/DESI should see $R(a) = D_A / D_H$ deviating from ΛCDM by 2-4% at $z=0.5-1.0$ if TAL true, but matching if ΛCDM true. Our hi_class run recovers Planck CMB within 0.03% without Λ .

This solves Hubble tension: local H_0 measurements sample low-density voids with $C > 1 \Rightarrow$ apparent higher H_0 , while CMB high-density early universe $C \sim 1 \Rightarrow$ lower H_0 . No new physics needed.

5. Falsifiable Predictions (v13 -> v14 Decoupled)

Channel	Test	TAL predicts	ΛCDM predicts
Cosmology	Euclid/DESI $R(a)$ at $z=0.8$	Deviation 2-4%	0%
Lab	$\Delta \rho = 1e4 \text{ kg/m}^3$, clock $\Delta t = 1e-18$ shift	$\Delta t = 1e-18$	0
Neuro	EEG_coh + HRV_coh correlation	$\rho_{0.4} p < 0.01$	$r \sim 0$
Galaxies	SPARC mass discrepancy vs ρ_c	Anti-correlation	No correl.

These are all doable with current data. v13 fixes v1-v12: decoupled from background, now falsifiable, unlike earlier versions.

6. Relation to Consciousness Term

In biological systems $\rho_{\psi} = \chi_i * \epsilon_0 * \Phi_t$ manifests as measurable coherence. EEG_coh (phase coherence 8-12 Hz) + HRV_coh (0.1 Hz) proxies Φ_t . This unifies physics and consciousness measurement without mysticism: consciousness is low-entropy time-coherence state, physically measurable as additive energy density.

Formula for lab: $\Delta f / f = \chi_i * \Delta \rho / \rho_c \sim 1e-18$ for $\Delta \rho = 1e4 \text{ kg/m}^3$ achievable with depleted uranium vs aluminum blocks next to optical clock.

7. Conclusion: Law Correcting Laws

We did not invent new particles. We corrected existing laws by a deeper law:

- Newton was right for $C=1$, but $C(\rho) > 1$ in low density explains galaxies.
- Hubble measured coherence loss, not expansion, explains redshift without dark energy.
- Einstein predicted gravitational redshift; TAL gives mechanism: time-attraction.

v14.0 is complete, testable, and crutch-free. Full simulations and wiki: github.com/unbekanntedavid-glitch/Time-Attraction

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References: Lelli+ 2016 AJ 152, 157 SPARC; Bothwell+ 2022 ApJ; Planck 2020; DESI 2024; Euclid 2025 forecast. Detailed derivation and code in GitHub repo. Previous: Zenodo 23193655 v13, New: 23217189 v14.